GreenGuard - Autonomous, Energy-Aware, Self-Healing LLM Inference System

Ishika Panwar
ishika.panwar.22cse@bmu.edu.in
220695

Komal Sharma
komal.sharma.22cse@bmu.edu.in
220716

ABSTRACT

This project addresses the unquantified and significant energy waste and system unreliability in Large Language Model (LLM) inference pipelines. Our solution, **GreenGuard**, is a comprehensive framework that uses a mixed-method approach to measure and mitigate these inefficiencies. We will empirically quantify **"Joules-per-token"** and **"gCO2e/token"** across diverse hardware and cloud environments, while also measuring the energy cost of system failures. The framework will then prototype and evaluate a suite of self-healing mechanisms (e.g., model swapping, automated restarts) to reduce this waste without affecting model accuracy or latency. Our key outcomes are a measurable reduction in LLM energy footprint and increased system uptime. The broader impact is the creation of a standardized AI Quality Assurance (AQa) Playbook to guide engineering teams in building more sustainable, reliable, and cost-effective AI deployments.

Keywords: Large Language Models (LLMs), Energy Efficiency, System Reliability, AI Operations (AIOps), LLM Inference, Carbon Footprint, Self-Healing Systems, AI Quality Assurance (AQa),Cloud Cost Optimization, Green AI, Self-Recovering AI Systems, Sustainable Computing

1. INTRODUCTION

In the past few years, we have witnessed a monumental leap forward in artificial intelligence, primarily driven by the advancements in Large Language Models (LLMs). These models, from GPT-4 to Llama 3, have transcended academic curiosity to become foundational technologies that power a vast array of applications, including search engines, customer service bots, code assistants, and creative platforms. This revolution, however, has an often-overlooked and critically important operational side: the immense energy footprint required to run these models at scale. Every single token generated, every query processed, consumes a tangible amount of electricity, and the sum of these millions of daily interactions results in a staggering energy demand. This consumption is not uniform; it's a highly dynamic variable influenced by a complex ecosystem of hardware, software, and geographical factors. The type of GPU (e.g., NVIDIA's A100 versus an H100), the underlying cloud platform (Azure, AWS, GCP), the geographic location of the data center, and the specific software optimizations applied to the model serving pipeline all contribute to a wildly fluctuating energy cost. This context underscores why the problem of energy-efficient and reliable LLM inference is no longer an academic abstraction but a pressing global challenge with profound economic, environmental, and operational implications.

The Core Problem

.The computational power required to train generative AI models that often have billions of parameters, such as OpenAI's GPT-4, can demand a staggering amount of electricity, which leads to increased carbon dioxide emissions and pressures on the electric gridThe core problem this project seeks to solve is the **unquantified and uncontrolled waste of energy and resources in LLM inference pipelines.**

- The current state of affairs is characterized by a "black box" approach to energy consumption. While companies can measure their total cloud spending, they often lack the granular data to understand the true energy costs on a per-query or per-token basis.

- This opacity prevents engineers from making informed, cost-effective, and environmentally conscious decisions.
- Compounding this issue is the inherent fragility of these complex systems. LLM inference pipelines are susceptible to a wide array of failures, from low-level issues like a GPU crashing or a memory leak to high-level problems like a bad model push.
- Each of these failures has a ripple effect: it not only disrupts service and leads to a poor user experience, but it also represents a significant waste of computational energy. The energy consumed by the request and the ongoing idle power draw of the failed system until it is manually or automatically restarted is pure waste.
- Our project aims to pull back the curtain on these hidden costs by providing a clear, data-driven understanding of how and where energy is being wasted. By defining and measuring metrics like "Joules-per-token" and "gCO2e/token," we can create a foundational layer for meaningful optimization and a more sustainable future for AI.

In this paper, we conduct the first comprehensive evaluation of power consumption and benchmark the energy efficiency across several widely used LLM inference engines, including vLLM, TensorRT-LLM, DeepSpeed, and Transformers. Our study provides a fine-grained analysis by decomposing the inference lifecycle into two stages: (1) the setup stage, which includes engine initialization and model loading steps; and (2) the token generation stage, where actual inference takes place. For each stage, we further measure real-time power consumption across key hardware components, including the total power input, GPU, CPU, and DRAM. Contributions of this paper include benchmarking several inference engines and presenting a breakdown analysis to share key insights about energy efficiency, and answering the following questions :

- During the setup stage, what is the power consump-tion of initializing inference engines and loading LLMs,until the first token is generated?
- During the token generation stage, how does power consumption and energy efficiency vary across infer-ence engines, considering GPU, CPU, and memory components?
- What is the relationship between energy efficiency and throughput? We investigate the hypothesis that higher throughput can improve energy efficiency by reducing per-token energy cost.
- Is there a single inference engine that optimizes energy efficiency across all scenarios?

BACKGROUND AND RELATED WORK

This section provides the background and reviews prior work relevant to our study, including the inference engines we evaluate and the related research on energy consumption during LLM inference.

Existing work in this area has primarily focused on isolated aspects of the problem. On the energy efficiency side, a significant body of research has explored hardware performance and the impact of different GPU architectures. For example, NVIDIA has published benchmarks on the performance-per-watt of their Hopper-series GPUs (H100) compared to the older Ampere-series (A100). Similarly, software optimizations like model quantization (converting models to lower-precision data types like INT8 or FP8), improved batching algorithms, and speculative decoding have been extensively studied to improve throughput and reduce latency, which indirectly leads to energy savings. However, these efforts often treat the systems as perfectly reliable, operating in a vacuum of perfect uptime. They fail to account for the real-world impact of failures, which can negate any gains made from these optimizations. On the reliability and fault tolerance side, engineering teams implement monitoring and alerting systems using tools like Prometheus and Grafana. These systems are invaluable for detecting failures, but they are fundamentally reactive. They inform an operator that a problem has occurred, but they do little to automatically correct the issue. This creates a reliance on manual intervention, which leads to significant downtime and, as we argue, a large amount of wasted energy.

GAP/NEED

This project identifies a critical **gap in the current LLM ecosystem**: the lack of an integrated framework that connects energy consumption, system reliability, and optimization strategies. There is no single "playbook" that helps engineering teams understand the complete picture of their operational costs—both economic and environmental. Our project fills this void by creating a holistic framework that simultaneously measures both sides of the problem: the energy cost of every successful token and the energy wasted due to every system failure. We believe that by creating a feedback loop between a system's energy consumption and its operational resilience, we can unlock a new class of optimizations that have been previously overlooked. By integrating energy telemetry with fault detection and autonomous recovery mechanisms, we can create a system that is not only faster and more reliable but also significantly more sustainable.

OBJECTIVE

The primary objectives of this project are to:

- **Empirically Measure Energy Footprint**: Accurately measure and compare the energy footprint of LLMs (Joules-per-token and gCO$_2$e/token) across different hardware (e.g., A100, H100) and cloud platforms under a variety of serving conditions.
- **Quantify Failure Costs**: Quantify the energy cost associated with a defined set of system failures, such as GPU overloads, memory leaks, and model-level errors.
- **Develop Self-Healing Mechanisms**: Develop and validate a suite of self-healing mechanisms designed to autonomously recover from these failures.
- **Perform Holistic Analysis**: Conduct a holistic analysis to determine which optimizations and self-healing mechanisms provide the greatest reduction in "Joules-per-token" without negatively impacting model accuracy or latency.
- **Create the AQa Playbook**: Synthesize all these findings into the GreenGuard AQa Playbook, a practical and actionable guide for building sustainable and resilient LLM pipelines for real-world deployments.

2. CONTRIBUTIONS

GreenGuard's primary contribution is the development of a unified framework that fundamentally re-engineers how large-scale LLM inference is managed. This is not merely an incremental improvement on existing tools; it represents a new paradigm that combines energy awareness, intelligent optimization, and autonomous recovery into a single, cohesive system.

Our novel contributions are:

1. A Holistic, Integrated Metrics System: We are pioneering a system that directly correlates energy consumption (measured as "Joules-per-token") with operational reliability. By precisely quantifying the energy waste caused by failures like GPU crashes, GreenGuard provides a single, holistic view of a pipeline's true cost, enabling intelligent, data-driven decisions that current fragmented monitoring tools simply cannot.

2. Autonomous, Self-Healing Intelligence: We contribute a suite of novel, self-healing algorithms that autonomously manage and optimize LLM pipelines. Unlike today's reactive systems that require manual intervention after a failure, GreenGuard can dynamically apply energy-saving optimizations and automatically recover from issues, ensuring continuous, efficient operation and maximizing resource utilization without human oversight.

3. The AI Quality Assurance (AQa) Playbook: The project's key practical contribution is the creation of a comprehensive, open-source AQa Playbook. This repeatable framework will serve as a definitive guide for deploying self-recovering and energy-efficient LLM systems. By democratizing these best practices, the playbook provides a clear, actionable roadmap for organizations to adopt sustainable AI, ensuring our research has a tangible, real-world impact.

4. Quantifying Waste from Failures: A specific contribution is the ability to not only identify system failures (like memory leaks) but to also precisely quantify the energy and carbon waste associated with them, providing a concrete metric for the value of our self-healing mechanisms.

5. Adaptive Optimization Algorithms: Our work includes the development of adaptive algorithms that can dynamically switch between different serving strategies (e.g., quantization, batching) based on real-time energy and performance metrics, optimizing for the lowest possible Joules-per-token without sacrificing accuracy or latency.

3. MOTIVATION

The environmental impact of AI has become a pressing global issue. Studies estimate that training and deploying large models can emit as much CO_2 as multiple car lifetimes. While training efficiency receives attention, inference—the stage where models are most widely used—remains an overlooked source of emissions. Every chatbot response, document summarization, or code completion request consumes electricity, and small inefficiencies multiply at scale.

Existing monitoring tools provide fragmented insights but lack integration and actionable feedback. CodeCarbon can estimate emissions, while Prometheus monitors GPU metrics—but developers still need to manually analyze the data. Furthermore, failures such as GPU crashes or memory leaks silently waste energy. In production settings, even a 1% failure rate can represent thousands of wasted GPU-hours annually.

Our team chose GreenGuard because it aligns with two critical goals: advancing **sustainable AI** and improving **operational reliability**. During research, we found little guidance for building inference systems that can adaptively optimize for energy or recover autonomously from failures. This gap inspired us to design a framework that bridges monitoring, optimization, and recovery.

The expected benefits are twofold: **environmental** and **economic**. By reducing Joules-per-token, organizations can lower both their cloud bills and their carbon footprint. By implementing self-healing mechanisms, they can ensure higher availability and reduce operational disruptions. GreenGuard is particularly relevant for businesses deploying LLMs at scale, academic researchers measuring AI's environmental cost, and policy makers interested in sustainable computing practices.

By making energy use and reliability metrics visible through an accessible dashboard and providing an actionable playbook, GreenGuard empowers stakeholders to make data-driven decisions for greener AI. This project reflects our belief that technological innovation should be paired with environmental responsibility, creating AI systems that are not only powerful but also sustainable.

Keywords: Environmental Impact of AI, Energy Efficiency, Carbon Footprint, Operational Reliability,

Keywords: GreenGuard Framework, Energy-Aware LLM Inference, Joules-per-Token, Self-Healing Systems, Adaptive Optimization, AI Quality Assurance (AQa), Sustainable AI